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Anillo polinomios

Proposición 1 (Anillo de polinomios). *Sea A un anillo (conmutativo con unidad)*

$$\implies A[x] := \{a_n x^n + \cdots + a_1 x + a_0 : a_i \in A \wedge n \in \mathbb{N}\}$$

es un anillo (conmutativo con unidad) con las siguientes operaciones:

- *Suma:* $(a_n x^n + \cdots + a_0) + (b_m x^m + \cdots + b_0) = (a_n + b_n)x^n + \cdots + (a_0 + b_0)$.
- *Producto:* $(a_n x^n + \cdots + a_0) \cdot (b_m x^m + \cdots + b_0) = \displaystyle\sum_{k=0}^{n+m} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k$.

Lo llamamos el anillo de polinomios en la variable x con coeficientes en A.

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